

HONGYUE JIN

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EDUCATION

Clemson University, Computing School, Clemson, SC
PhD Program

January 2024 - Now

Tufts University, School of Engineering, Medford MA
Master of Science, Electrical Engineering (GPA:3.75)

September 2021 - May 2023

Southeast University, Nanjing, China
Bachelor of Technology, Information Engineering (GPA:3.19)

September 2016 - June 2020

RELEVANT COURSEWORK

Parallel Computing, Computer Engineering, Communication System, Digital Signal Process, Network Security

TECHNICAL SKILLS

Technology/Techniques/Tools:

Linux, MATLAB, LaTeX, Vivado

Programming Languages:

C++, Cuda, Python, MATLAB, VHDL, Assembly Language

APPLIED ACADEMIC PROJECTS

Clemson University, Clemson, SC

Jan 2024 – Now

Research Project: Side-Channel Vulnerabilities in Virtualized GPUs (Accepted by NDSS 2026)

- Conducted a year-long research project on side-channel vulnerabilities in virtualized NVIDIA GPUs in cloud environments. Designed and implemented a tailored vulkan based Prime+Probe attack primitive for TLBs, overcoming GPU virtualization challenges such as memory constraints and time-sharing execution.
- Conducted a year-long research project investigating side-channel vulnerabilities in virtualized NVIDIA GPUs used in cloud environments.
- Designed and implemented a customized \texttt{Prime+Probe} attack primitive targeting TLBs, addressing unique challenges in vGPU setups, including memory constraints, time-sharing execution, and synchronization issues.
- Overcame vGPU memory management limitations by leveraging Vulkan APIs to allocate 64KB pages, enabling precise manipulation of GPU TLB states and reliable eviction set construction.
- Orchestrated efficient priming and probing operations across multiple TLB sets, ensuring minimal interference and accurate contention measurement in the time-sharing GPU execution model.
- Demonstrated practical attacks through two case studies:
 - A cheating exploit in Counter-Strike 2 that revealed hidden opponents by analyzing GPU TLB access patterns.
 - A website fingerprinting attack achieving up to 90% accuracy in identifying web pages browsed on virtual desktops.

Tufts University, Medford, MA

September 2021 – MAY 2023

Research: Exploration on HotGauge using fine-grained DFS

- Collaborated on the research project "HotGauge," aimed at developing a methodology for mitigating advanced hotspots in modern and next-generation processors.
- Contributed to the investigation of advanced thermal hotspot mitigation strategies using the HotGauge tool.
- Played an integral role in devising innovative solutions by leveraging work steering and Dynamic Frequency Scaling (DFS) techniques.
- Actively participated in the successful implementation of fine-grained DFS within HotGauge and conducted comprehensive experiments to validate its efficacy.
- The results conclusively confirm the effectiveness of DFS in mitigating advanced thermal hotspots.

Project: Verifying Thermal Side Channels of Encryption Algorithms on FPGA, Course: **Advanced Topics in Computer Architecture** (EE 156)

- Led a teamwork project to investigate thermal characteristics of encryption algorithms and proposed algorithm-level parameters for measurement.
- Extracted thermal feature parameters from FPGA experiments, achieving 90.4% accuracy in identifying algorithms

using an artificial neural network.

- Findings contribute to understanding and protecting FPGA encryption algorithms against thermal side channels.

Project: Implementing a 5-stage ARM processor in VHDL, Course: **Computer Engineering** (EE 126)

- Led an independent 8-week course project to implement a modified version of a 5-stage Pipelined CPU capable of executing a subset of ARM v8-64bit ISA using VHDL.
- Overcame data hazards using forwarding and stalling techniques, ensuring efficient data handling within the pipeline.
- Addressed control hazards by resolving conditional and unconditional branches in the Instruction Decode (ID) stage and implementing flushing mechanisms to maintain pipeline efficiency.

Project: Shared Memory Optimization for CNN Computation on GPUs, Course: **Parallel Computing** (EE 155)

- Developed a project centered on computing a Convolutional Neural Network (CNN) using C++ and CUDA, with a focus on addressing edge effects.
- Implemented an innovative algorithm that eliminated the need for a critical section, utilizing shared memory to optimize performance and avoid repeated access to device memory.
- Applied the CNN to matrices of sizes 4Kx4K and 8Kx8K, using filter sizes of 4x4, 8x8, and 12x12.

Project: Simulation of Modern Communication Systems, Course: **Communication Systems** (EE 107)

- Developed an independent 8-week course project to simulate a digital communication system, transmitting and receiving images over a noisy channel, mirroring real communication systems.
- Implemented MATLAB to create essential system blocks, such as the Image Processing, Modulation, Channel, Matched Filter, Receiver Equalization, and Sampling & Detection blocks.
- Executed four transmission modes to transmit '.jpg' format pictures, conducting detailed comparisons among them and documenting findings in a comprehensive paper.

Southeast University, Nanjing, China

Research: Research on Network Device Identity Detection Based on Active Search Mechanism 2020

- Conducted a 20-week individual graduation project to propose a network device detection system based on active search mechanisms, sending simulated impulse functions to clients and collecting their impulse response data characteristics.
- Developed a scheme utilizing the ICMP protocol to obtain the delay from the host to the target network device.
- Simulated the probe as an impulse response by designing the day part of the probe and obtained the impulse response of the target network device.
- Applied a fully connected artificial neural network algorithm for system identity identification.
- Constructed a prototype system for active discovery mechanisms of network devices, outlining its four modules: probe generation, probe sending and receiving, data feature extraction, and identity authentication.
- Collected and analyzed information from over 200,000 pings on 10 target network devices, achieving a high accuracy rate of 84% in identification.

CERTIFICATES

Honorable Mention of Interdisciplinary Contest in Modeling	January 2019
Honor of Merit Student in 2016 & 2019	October 2019
Graduation Certificate of Honor by Chien-Shiung Wu college of Southeast University	July 2020

CAMPUS ENGAGEMENT

- Member of the student union, responsible for managing the activity room of the college
- Member of the debating team of the college
- Member of the Organization Department of the Youth League Committee of the College
- A national second-class player of chess, alternate chess association master and national primary coach; winning the 10th place and honor of Alternate chess master at the Hebei International Chess Tournament